

Results (Predictions)

Dataset:

```
      objid      u      g      r      i      z  \
0  1237671696060581716  23.67549  20.66940  19.22081  18.22113  17.70474
1  1237668707303424749  21.43174  19.40755  18.49240  18.13505  17.94761
2  1237668706229485848  21.85019  20.34860  19.69839  19.45158  19.40700
3  1237668705155613223  22.01424  20.00161  19.17358  18.85867  18.58247
4  1237668706766422345  17.68311  16.56997  16.18189  16.04864  15.99897

      redshift
0  0.000470
1  0.000034
2  0.000053
3  0.000284
4  0.000233
Index(['objid', 'u', 'g', 'r', 'i', 'z', 'redshift'], dtype='object')
```

XGB Classifier (Higher order Indices with Redshift)

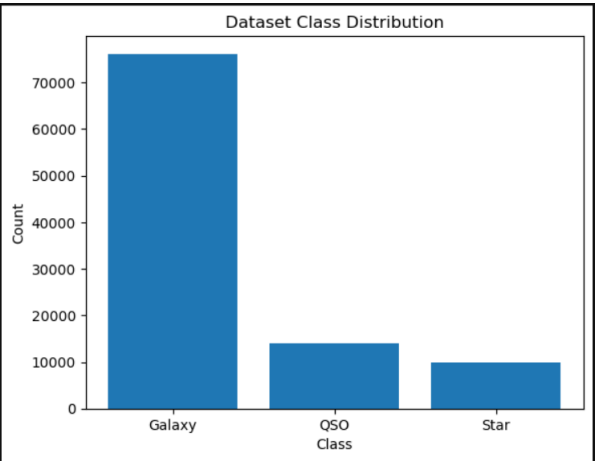
Confidence:

	predicted_class	confidence
0	Star	0.998103
1	Star	0.999910
2	Star	0.999885
3	Star	0.998901
4	Star	0.999849
5	Star	0.999947
6	Star	0.999959
7	Star	0.999715
8	Star	0.999796
9	Galaxy	0.999667

Value Counts:

predicted_class	
Galaxy	76156
QSO	14046
Star	9798
Name: count, dtype: int64	

Dataset Class Distribution:



XGB Classifier (Higher order Indices without Redshift Prediction)

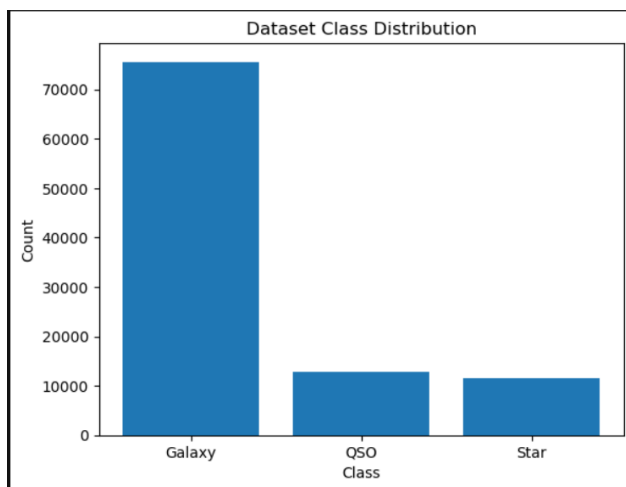
Confidence:

	predicted_class	confidence
0	Star	0.965547
1	Star	0.982043
2	Star	0.987484
3	Star	0.979906
4	Star	0.997320
5	Star	0.990591
6	Star	0.995285
7	Star	0.574337
8	Star	0.961295
9	Galaxy	0.877086

Value Counts:

```
predicted_class
Galaxy      75571
QSO         12929
Star        11500
Name: count, dtype: int64
```

Dataset Class Distribution:

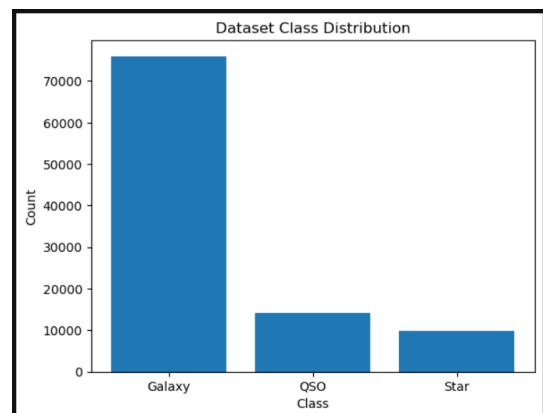


FNN Classifier (Higher order Indices with Redshift Prediction)

Confidence and Value Counts

	predicted_class	confidence
0	Star	0.999979
1	Star	0.999245
2	Star	0.997567
3	Star	0.998018
4	Star	0.999259
5	Star	0.999463
6	Star	0.999153
7	Star	0.999955
8	Star	0.999999
9	Galaxy	0.978944
predicted_class		
	Galaxy	75994
	QSO	14184
	Star	9822
Name: count, dtype: int64		

Dataset Class Distribution

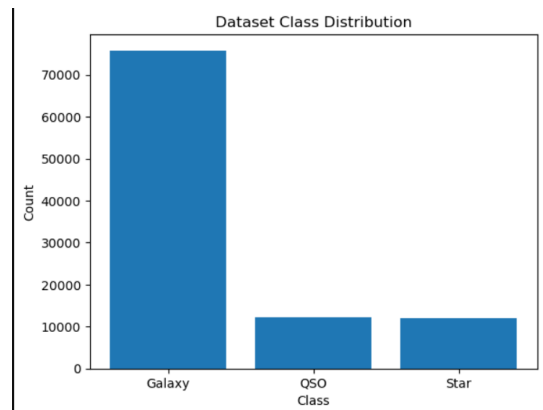


FNN Classifier (Higher order Indices without Redshift Prediction)

Confidence and Value Counts

```
predicted_class confidence_no_z
0             Star      0.924642
1             Star      0.998584
2             Star      0.997462
3             Star      0.987626
4             Star      0.998718
5             Star      0.991356
6             Star      0.993312
7             Star      0.843181
8             Star      0.927894
9             Galaxy     0.997535
predicted_class
Galaxy      75814
QSO         12240
Star        11946
Name: count, dtype: int64
```

Dataset Class Distribution



Stacking (FNN + XGB) (with redshift (prediction))

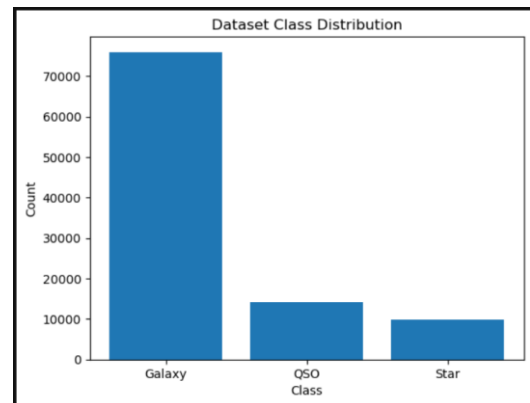
Confidence and Value Counts

```
predicted_class confidence
0             Star      0.997619
1             Star      0.997636
2             Star      0.997633
3             Star      0.997630
4             Star      0.997636
5             Star      0.997637
6             Star      0.997638
7             Star      0.997631
8             Star      0.997635
9             Galaxy     0.995329

Class distribution:
predicted_class
Galaxy      75925
QSO         14259
Star         9816
Name: count, dtype: int64

High confidence count: 98798
```

Dataset Class Distribution



Stacking (FNN + XGB) (without redshift (prediction))

Confidence and Value Counts

	predicted_class_no_z	confidence_no_z
0	Star	0.974441
1	Star	0.975521
2	Star	0.975484
3	Star	0.975300
4	Star	0.976722
5	Star	0.976247
6	Star	0.976506
7	Star	0.719816
8	Star	0.970641
9	Galaxy	0.970335

Class distribution:

predicted_class_no_z

Galaxy 74677

QSO 12842

Star 12481

Name: count, dtype: int64

Dataset Class Distribution

